Electrical Science
ENGR 2303

Video 3

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Series and Parallel Resistors

- Series Resistors and Voltage Division.
- Parallel Resistors and Current Division.
- Wye-Delta Transformations.
- Examples

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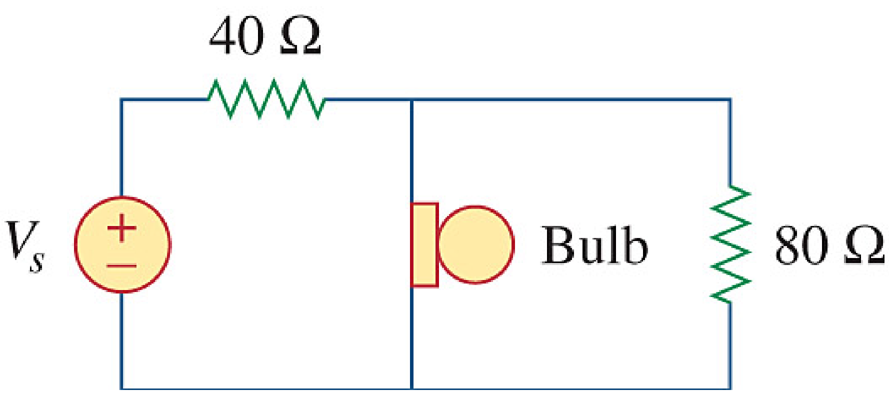
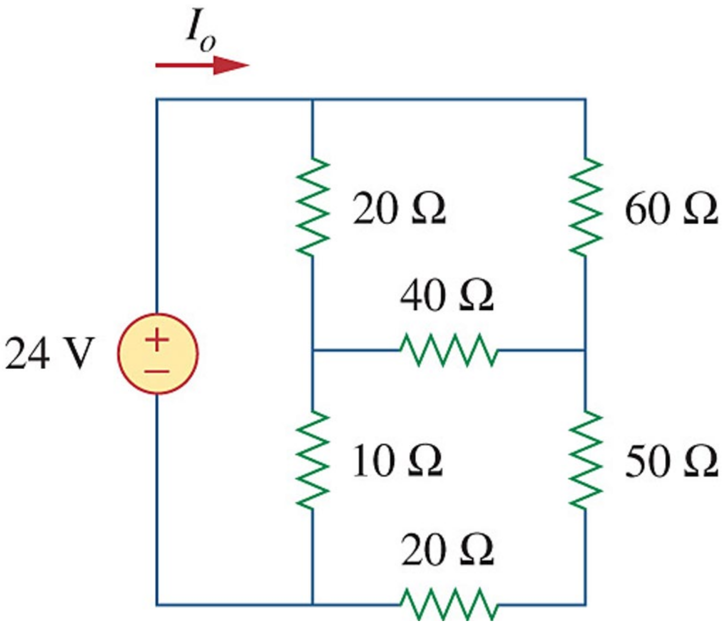




Circuit Theory Video 3

Series Resistors and Voltage Division

- Series: Two or more elements are in series if they are cascaded or connected sequentially and consequently carry the same current.

- The equivalent resistance of any number of resistors connected in a series is the sum of the individual resistances.

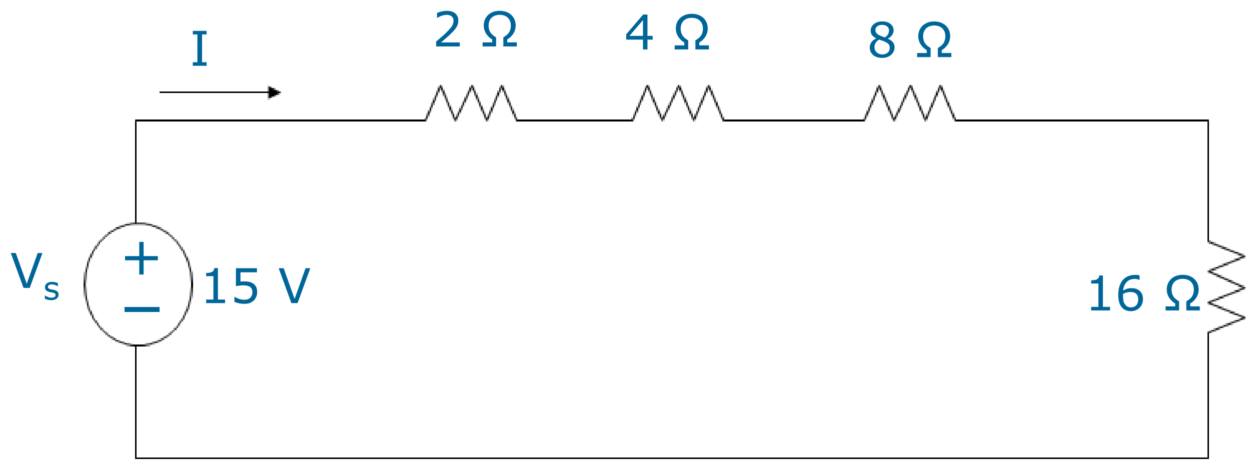
$$R_{eq} = R_1 + R_2 + \cdots + R_N = \sum_{n=1}^{N} R_n$$

- The voltage divider can be expressed as

$$v_n = \frac{R_n}{R_1 + R_2 + \cdots + R_N} v$$

Series Resistors and Voltage Division

Example: Find the current I and the voltage across the 16Ω resistor.



Parallel Resistors and Current Division

- Parallel: Two or more elements are in parallel if they are connected to the same two nodes and consequently have the same voltage across them.

- The equivalent resistance of a circuit with N resistors in parallel is:

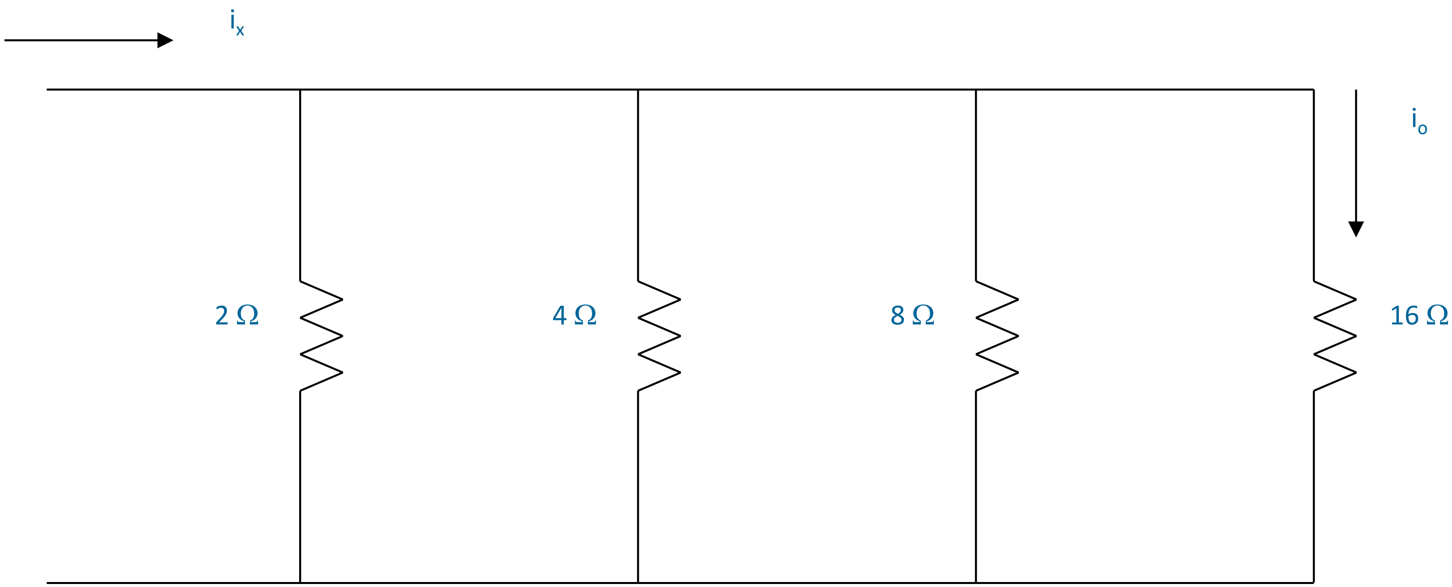
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_N}$$

- The total current i is shared by the resistors in inverse proportion to their resistances. The current divider can be expressed as:

$$i_n = \frac{v}{R_n} = \frac{iR_{eq}}{R_n}$$

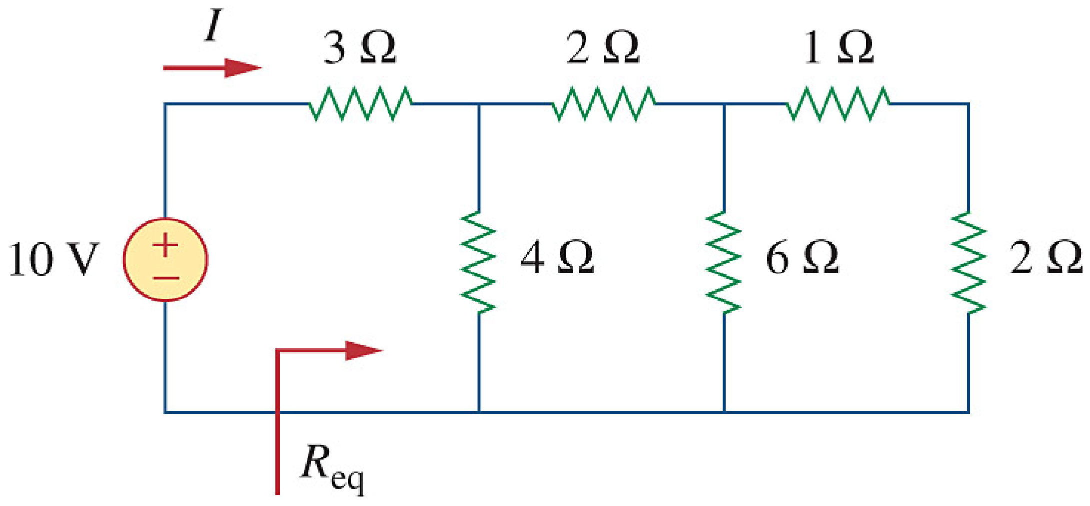
Parallel Resistors and Current Division

<u>Example</u>: For the circuit below, i_o =2 A. Calculate i_x and the total power dissipated by the circuit.

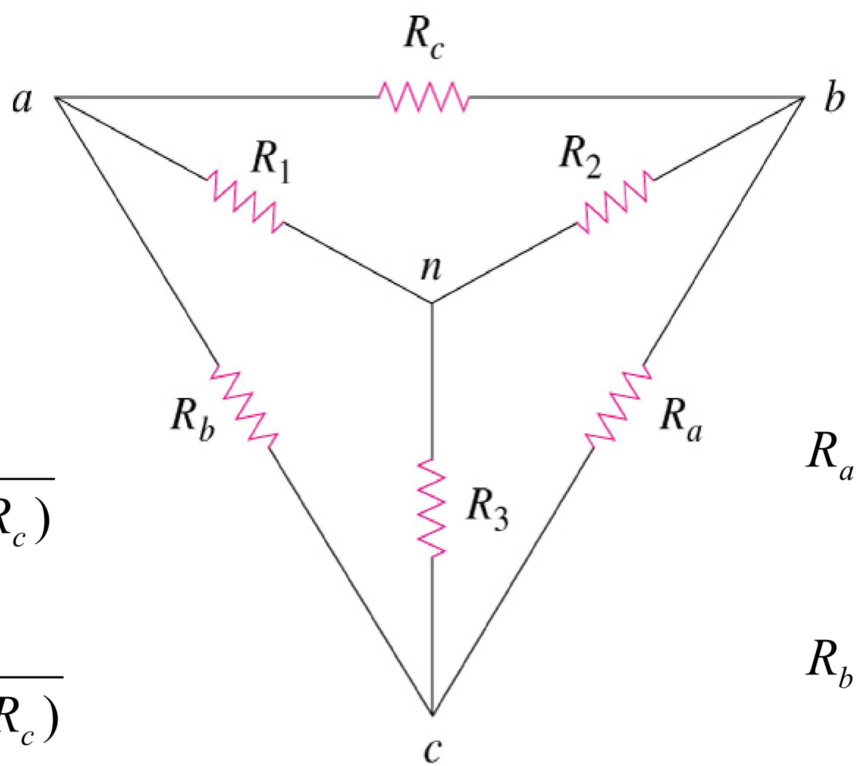


Parallel Resistors and Current Division

- For the ladder network below, find I and R_{eq}.



Wye-Delta Transformations



Delta -> Star

$$R_1 = \frac{R_b R_c}{(R_a + R_b + R_c)}$$

$$R_2 = \frac{R_c R_a}{(R_a + R_b + R_c)}$$

$$R_3 = \frac{R_a R_b}{(R_a + R_b + R_c)}$$

Star -> Delta

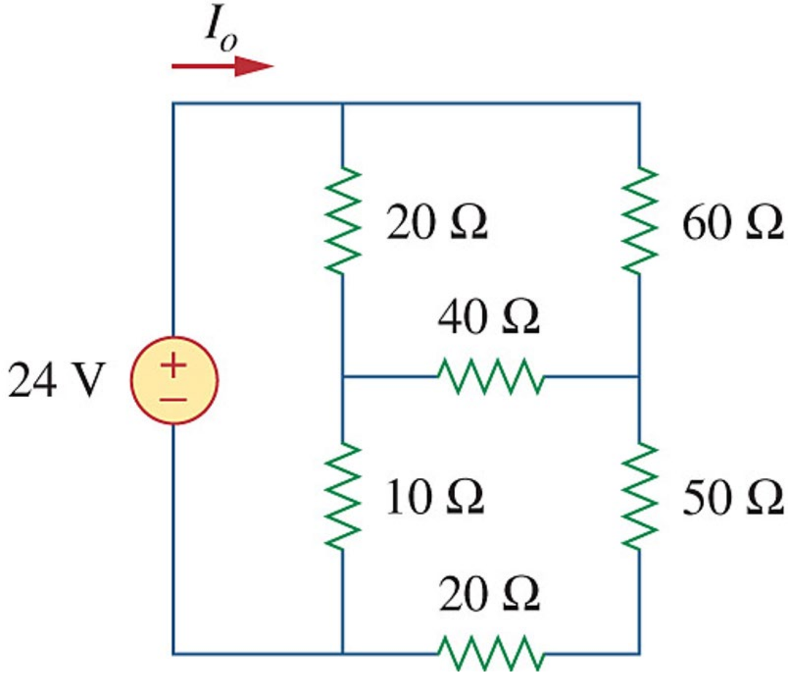
$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}$$

$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2}$$

$$R_c = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3}$$

Wye-Delta Transformations

Calculate I_o in the circuit



Application Problem

- The lightbulb is rated 120 V, 0.75 A. Calculate V_s to make the lightbulb operate at the rated conditions.